Creep

Creep is a slow and gradual movement of soil downhill. Its velocity is typically less than a centimetre per year. Freezing and thawing contribute the soil creep by progressively moving soil particles down the hill. Creep is manifested at the surface by things like tilted utility poles, fences and trees Vegetation helps reduce the rate of soil creep.

4.5. Gradational Processes

Gradation is the process by which the earth's surface gets leveled. It can be further divided into degradation, the process of eroding the earth's elevated surface and aggradations, the process of filling up the earth's depressions.

4.5.1 Gradational Agents

The forces which act on the surface of the earth are termed as Gradational agents. Water, wave, wind, ice are the important gradational agents. Let us now discuss the gradiational agents one by one.

4.6. The River

The streams have a huge capacity to erode the rock over which they flow. In fact, the formation of the river channel is the result of the erosional capacity of the stream. The erosional capacity of the stream depends on its volume of water and velocity of flow. The river performs three types of work. They are erosion, transportation and deposition.

1. **Erosion**: The breaking of rocks by the river in along its course is called erosion. Erosional work of a river is performed mechanically and chemically. River erosion is carried out in the following ways:
   i. **Hydraulic action**: It refers to the physical force of the moving water which breaks the rocks in its course.
   ii. **Corrasion (abrasion**): It refers to the breaking of rock in the bed and on the bank by fragments carried by the stream.
   iii. **Corrosion(solution**): It refers to the dissolving process of soluble minerals by the splashing of stream water.
   iv. **Attrition**: It refers to the eroded materials carried by the stream strike against each other.
2. **Transportation**: Stream carrying the fragmented materials broken by the stream is called transportation. After erosion, the eroded materials get transported along with the running water. This transportation of eroded materials is carried in four ways:
   i. **Traction**: The heavier and larger rock fragments like gravels, pebbles etc are forced by the flow of the river to roll along its bed. These fragments can be seen rolling, slipping, bumping and being dragged. This process is called as traction and the load transported in this way are called traction load.
   ii. **Saltation**: Some of the fragments of the rocks move along the bed of a stream by bouncing continuously. This process is called as saltation.
   iii. **Suspension**: The holding up of small particles of sand, silt, and mud by the water as the stream flows is called suspension.
   iv. **Solution**: Some parts of the rock fragments dissolve in the river water and transported. This type of transportation is called solution transportation.
3. **Deposition**: When the velocity of the stream decreases, the stream deposits sand, silt and other fragments. It is

TN_Govt_Geography_XI_Std Unit_04.indd 84 2/3/2020 4:24:11 PM